MIL-C-48129(PA)
9 Jan 1975

MILITARY SPECIFICATION

CHARGE, PROPELLING, XM195
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing of XM195 Propelling Charges for 81MM mortar ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 -Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 -Quality Assurance Terms and Definitions
MIL-STD-1168 -Lot Numbering of Ammunition
MIL-STD-1235 -Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PRODUCT AND PACKING DRAWINGS

ARMY

9257039 -Charge, Propelling, XM195

FSC: 1315

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9281064 -Packing and Marking for Box, Fiberboard, Packaging, Re-usable, Collapsible, for Support Increment with Charge, Propelling, XM195

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

49 CFR 71-90 Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles

(Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-90 (latest revision)."

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The XM195 Propelling Charge shall comply with all requirements specified on Drawing (dwg.) 9257039 and with all requirements specified in applicable specifications.

3.3 Gross Weight of Propellant Increment (see dwg. 9257041).-The weight of the loaded propellant increment within each lot shall not vary more than plus four or minus four grains from the established propellant increment weight (assessed propellant weight plus propellant bag tare weight-see 4.2.4.2).

3.4 Leak Test (see dwg. 9257039).-The loaded propelling charges shall show no evidence of leaking when tested as specified in 4.3.1.

3.5 First Article Inspection.-A first article sample shall be submitted in accordance with para. 4.2.2. Samples shall be chosen, tested and observed for compliance with requirements in accordance with this specification.

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3.6 Residue Test.-First article samples shall be tested in accordance with para. 4.2.4.4. to determine compliance with the Residue Test requirements specified therein.

3.7 Velocity.-The propelling charges shall impart a corrected muzzle velocity of not more than 905 feet per second (fps) and not less than 875 fps when tested as specified in 4.3.3. In addition, the standard deviation of the lot shall be not greater than 4.0 fps.

3.8 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of dirt, grease and other foreign material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government of commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Nuber of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted

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shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Quality Conformance Inspection

4.2.1 Lot Formation.-The term "inspection lot" as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, perfomance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Increment Container parts of one lot interfix from one manufacturer.
b. Propellant from not more than one lot.
c. Propellant Increments from one lot interfix from one manufacturer.
d. Adhesive from not more than one manufacturer.

4.2.2 First article inspection

4.2.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.2.2.2. The first article sample shall be selected from one lot of propelling charges which has been produced by the contractor and has been manufactured by the same production process, using procedures and equipment which will be used in fulfilling the contract. All materials including the packaging and packing shall be obtained from the

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same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects shall not be submitted unless specifically designated by the contracting officer. A first article sample shall also be submitted as directed by the contracting officer, whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in the manufacturing process, materials used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

a. Fifty charges shall be submitted to a Government laboratory for examinations and tests specified under 4.2.3 and 4.2.4.1 through 4.2.4.3.

b. Seven hundred charges shall be submitted to a Government proving ground for residue test, 4.2.4.4 and velocity test, 4.2.4.5.1.

4.2.2.2 Rejection.-If the propelling charges fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.2.3 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235, may be used if approved by the Government. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.5.

4.2.3.1 Bag, Propellant (see dwg. 9257040 covering a detail of dwg. 9257039).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
101.	Length	Gage (CD)
102.	Width	Gage (CD)

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Minor:
201. Improper stitch.....Visual
202. Improper number of stitches.....Visual
203. Marking missing, misleading or unidentifiable.....Visual
204. Evidence of poor workmanship(see 3.8).Visual

4.2.3.2 Increment, Propellant (see dwg. 9257041 covering a detail of dwg. 9257039).

Categories	Defects	Method Of Inspection
Critical:		
	1. Bag or bag seam damaged or stitches missing to extent that propellant can escape	Visual
Major:	None defined.	
Minor:	None defined.	

4.2.3.3 Charge, Propelling, XM195 (see dwg. 9257039).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
	101. Height	Gage (CD)
	102. Outside diameter	Gage (CD)
	103. Length	Gage (CD)
	104. Inside diameter	Gage (CD)
	105. Width of opening	Gage (CD)
Minor:		
	201. Taper of opening	Gage (CD)
	202. Marking missing, misleading or unidentifiable	Visual
	203. Evidence of poor workmanship(see 3.8)	Visual

4.2.4 Testing

4.2.4.1 Weight of propellant-Critical Defect (under minimum weight); Major Defect (over maximum weight).-Prior to loading, the propellant shall be weighed 100 percent. Equipment shall be CD (see 6.2).

4.2.4.2 Gross weight of propellant increment (see 3.3)-Critical Defect (under minimum weight); Major Defect (over maximum weight).-The loaded propellant increments

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shall be weighed 100 percent. Any propellant increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be CD (see 6.2).

4.2.4.3 Leak Test (see 3.4) Critical Defect.-The propelling charges shall be tested 100 percent. Any propelling charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.1. Equipment shall be CD (see 6.2).

4.2.4.4 Residue Test (see 6.3) Special Defect.-The contractor shall submit a first article sample of five hundred (500) XM195 Propelling Charges for evaluation in accordance with the provisions of 4.3.2. The samples shall be selected from one lot of propelling charges, from the production process and using procedures and equipment which will be used in fulfilling the contract. The samples shall be inspected to assure that they conform to dwg. 9257039 and inspection requirements of this specification. Defective samples will not be submitted.

4.2.4.5 Velocity (see 3.7) Major Defect.-

4.2.4.5.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.81 exceeds the lot requirement, the lot shall be rejected. This test shall be performed as specified in 4.3.3.

4.2.4.5.2 After three consecutive lots have met the criteria of 4.2.4.5.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter I, and an AQL of 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.75 exceeds the applicable lot requirement, the lot shall be rejected. This test shall be performed as specified in 4.3.3.

4.2.5 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.3 and 4.2.4 (see 6.2).

4.3 Test Methods and Procedures

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4.3.1 Leak Test.-The propelling charges shall be conditioned to and tested at a temperature of 70 to 80 degrees Fahrenheit (°F). Propelling charges shall be assembled onto a 1.102-.002 inch diameter rigid bar. Propelling charges shall be subjected to a surrounding absolute pressure of 15 to 10 inches of mercury vacuum. Vacuum shall be drawn gradually and uniformly with respect to time. Vacuum shall be attained in no less than ten seconds. While the vacuum is being maintained, an approximate ten second observation shall be made for leakage as evidenced by the lack of container billowing. Evidence of leakage shall be cause for rejection.

4.3.2 Residue Test.-This test shall be conducted at a Government proving ground. Test rounds (81MM-M374A3 mortar rounds) shall be assembled with first article sample XM195 propelling charges. Reference rounds shall be those selected from current production (81MM-M374A2 mortar rounds) at the time of first article testing.

Two hundred test rounds shall be test fired. These rounds shall consist of four fifty (50) round groups. Each group shall be assembled at a different charge zone. The charge zones shall consist of charges one, two, three and four. Two hundred reference rounds shall be available for firing. These rounds shall consist of four fifty (50) round groups consisting of charge zones one, three, seven and nine.

The rounds shall be conditioned to and fired at 70 degrees Fahrenheit. All rounds shall be fired at an elevation of 800 mils, from an 81MM mortar tube (3.207 inches max.) for which the round is standard, at a rate of 12 rounds per minute. The mortar tube shall be cooled between the firing of each group. In addition, the tube shall be dry swabbed and elevation checked between the firing of each group. Observance shall be made for misfires.

The first article sample will be considered acceptable if no misfires occur. It will then be unnecessary to test the reference rounds. However, if a misfire occurs, it shall be determined if it was caused by residue from the propelling charge. Mortar tube elevation shall be checked. A misfire caused by residue will constitute rejection of the first article sample. In this event the reference rounds will be tested without delay, at the completion of the firing of that test round group, at that charge zone corresponding to that at which the test round misfired.

4.3.3 Velocity.-This test shall be conducted at a Government proving ground. The rounds shall be inert loaded to 9.05 plus or minus .02 pounds including the weight of the dummy fuze. The rounds shall be assembled at maximum charge (i.e. with four charges). The rounds shall be

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conditioned to 70 degrees Fahrenheit and fired at an elevation of 800 mils, from an 81MM mortar tube (3.207 inches max.) for which the round is standard. Reference rounds shall be fired using reference standard components and a calibration lot of propellant charges and ignition cartridges for velocity correction. The individual velocities and pressures shall be obtained and recorded. Velocity standard deviation and mean velocity shall be calculated to determine compliance with applicable requirements. A formal firing report, with this data and lot acceptability, shall be sent to Commander, Picatinny Arsenal, Dover, New Jersey (see 6.4).

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with drawing (dwg.) 9281064 and Level C shipping.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each inspection lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet.

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When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable specific identifying information from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Special defects

6.3.1 Definition.-A special defect is a defect that judgement and experience indicate could, depending on the degree of variance from the design requirement, result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

6.3.2 Disposition.-Upon investigation and analysis of lots exhibiting special defects, by the responsible technical activity, a recommendation of acceptance or rejection shall be made to Director of Quality Assurance, U.S. Army Armament

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Command. Fin al acceptance of such lots on waiver shall be made only by the Director of Quality Assurance, U.S. Army Armament Command.

6.4 Submission of data.-One copy of proving ground test data (i.e., firing record containing Residue & Velocity Test results) shall be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A, SARPA-AD-E-A4
Dover, New Jersey 07801

6.5 Proving Ground test summary.

TEST	SAMPLE SIZE	REQUIREMENTS
First Article Inspection	(see 4.2.4.4&4.2.4.5) Five Hundred (500) Two Hundred (200)	a. Residue Test (see 3.6) b. Velocity (see 3.7)
Velocity (normal production)	One Hundred (100)	a. See para. 3.7

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1315-A002

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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